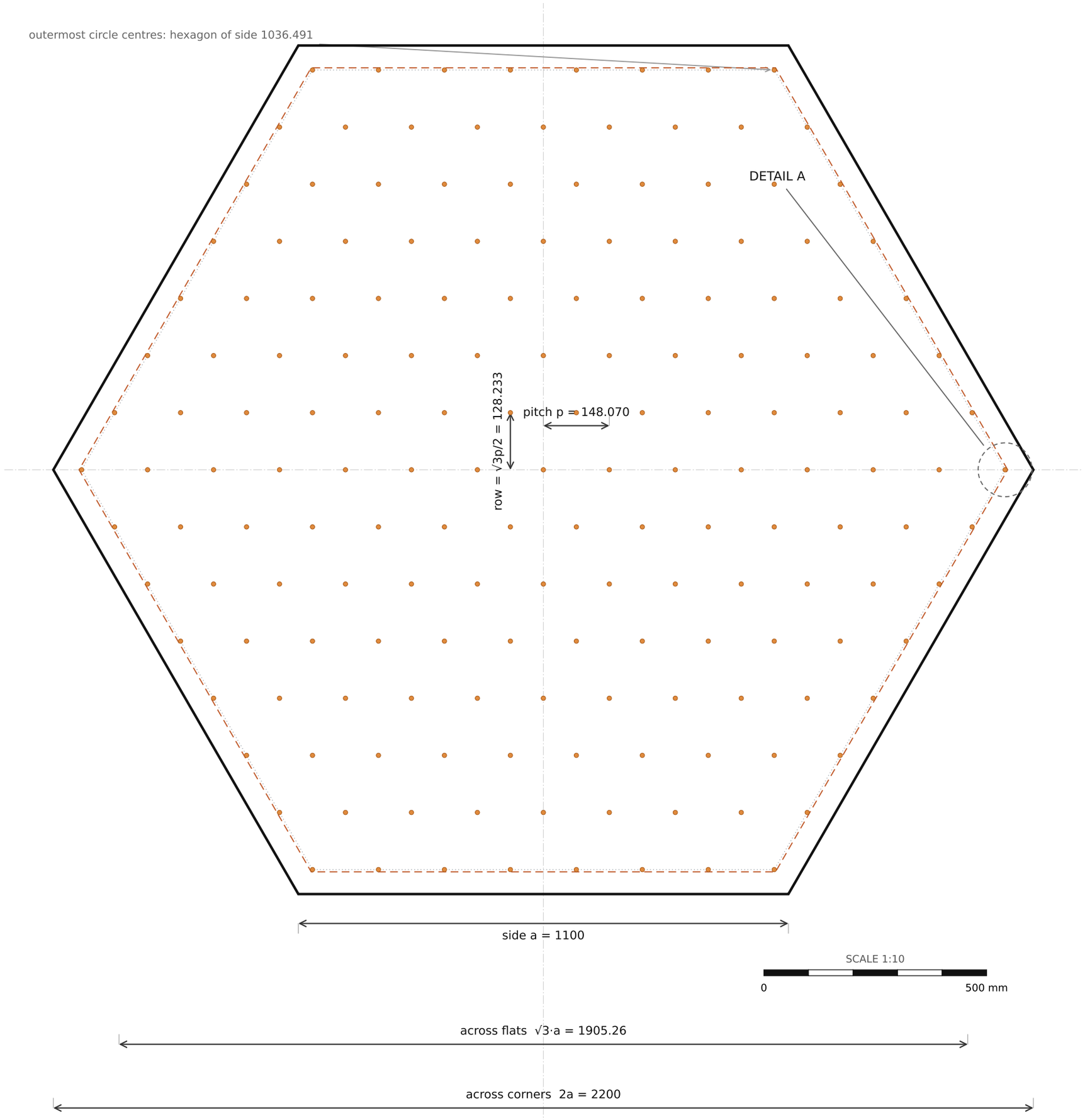

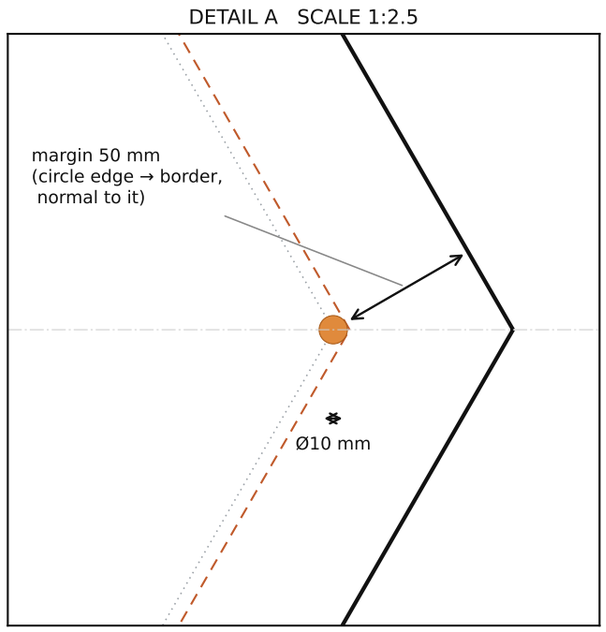

NOTES

1. N = 169 circles on a triangular (hexagonal) lattice, 7 complete shells about the centre, 6-fold symmetric.
2. MARGIN 50 mm = clear distance from the outermost circle EDGE to the hexagon border, measured normal to the border. Corner circles govern; see DETAIL A.
3. Circle Ø10 mm drawn at true size. No exaggeration anywhere on this sheet.
4. Lattice: a1 = (148.0702, 0), a2 = (74.0351, 128.2326) mm.
5. Shell radii: k·p, k = 1..7 (outermost 1036.4915 mm).
6. All dimensions in millimetres.

HEXAGONAL CIRCLE ARRAY	N = 169	SCALE 1:10 (sheet 14.33 x 12.57 in)
hexagon side a = 1100 mm	across flats = 1905.256 mm	across corners = 2200 mm
circle Ø = 10 mm (r = 5)	margin to border = 50 mm	centre locus hexagon side = 1036.4915 mm
pitch p = 148.0702 mm	row spacing = 128.2326 mm	clear gap between circles = 138.0702 mm
shells k = 7	total circle area = 13273.2290 mm²	hexagon area = 3143672.2 mm² (fill 0.42222 %)